



Electromagnetic Simulation Software

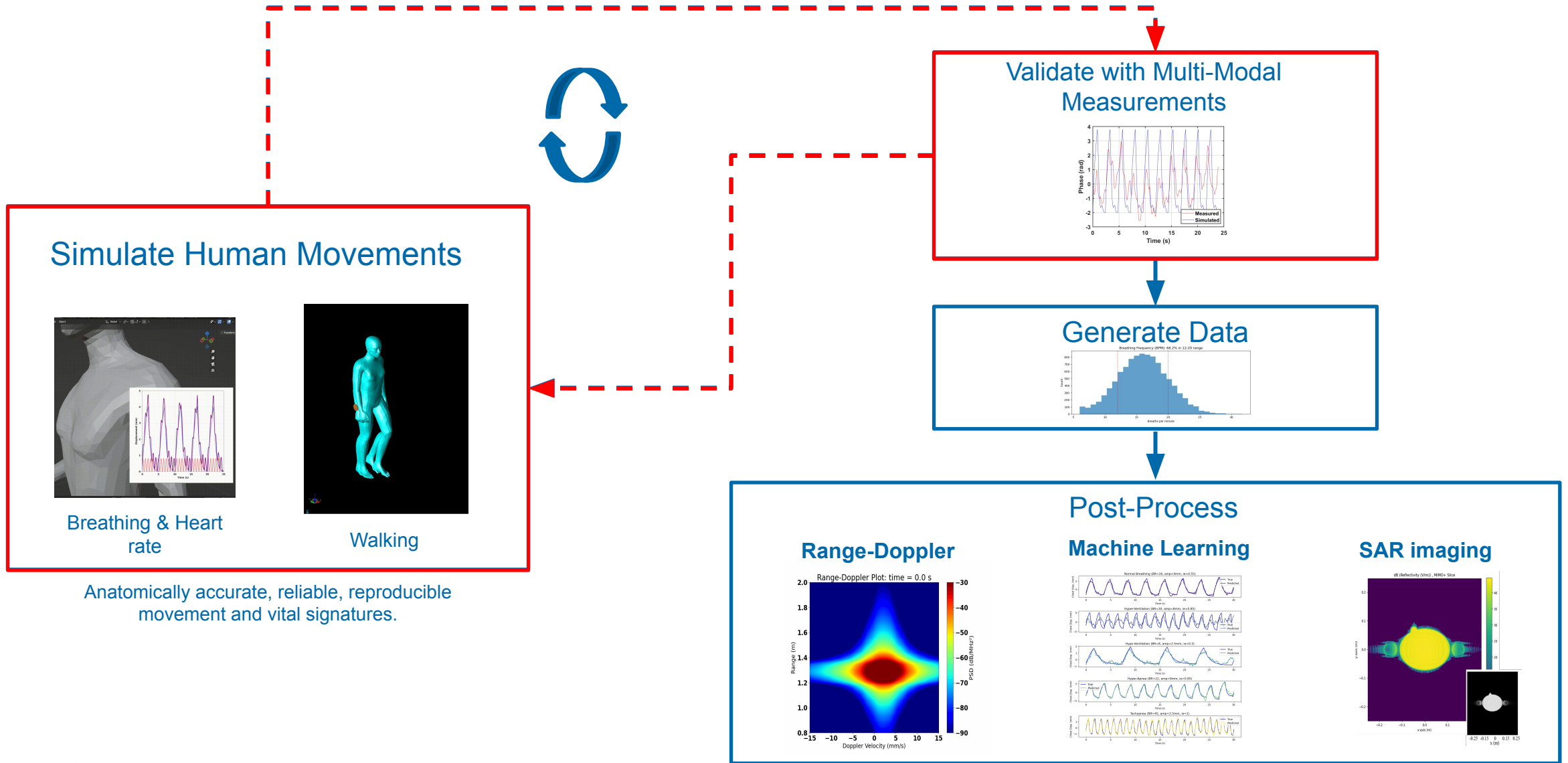
Synthetic Radar Return-Based Neural Network for Detecting Breathing Anomaly

Ben Hardy, Swagato Mukherjee, and Tarun Chawla

Remcom
ISAP 2025 福岡 (Fukuoka), Japan



Outline

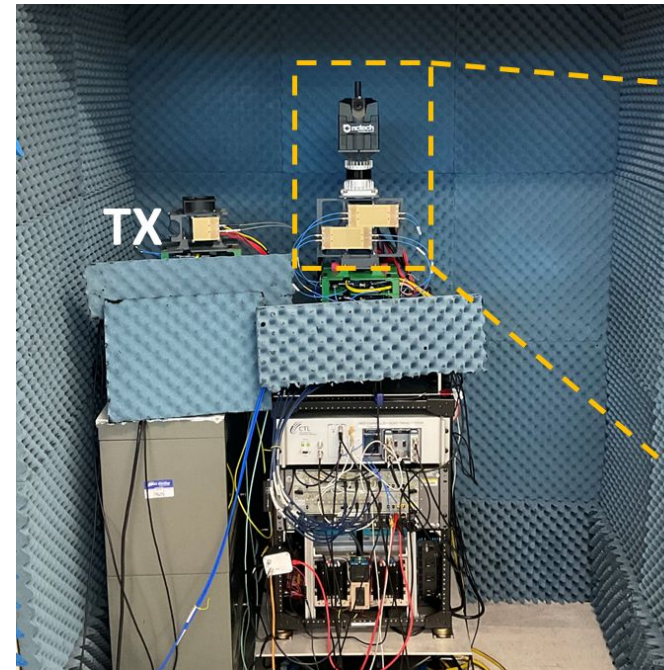


NIST Measurements

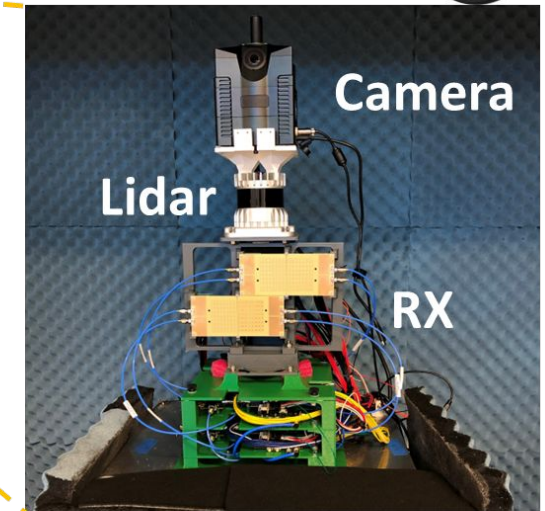
Courtesy of



- 1 dual-polarized TX phased-array antenna board (64 microstrip antennas)
- 4 dual-polarized RX phased-array antenna boards (256 microstrip antennas)
- 360° (AZ) camera, 360° (AZ) lidar, vitals sensor



Vital sign monitoring



Simulating Anatomical Movements

$$r_{breath}(t) = \frac{-r_{max}}{T_i T_e} t^2 + \frac{r_{max} T}{T_i T_e} t \text{ for } t \leq T_i$$

$$r_{breath}(t) = \frac{r_{max}}{1 - e^{-T_e/\tau}} \left(e^{-(t-T_e)/\tau} - e^{-T_e/\tau} \right) \text{ for } T_i < t \leq T$$

T_i – Inspiration Time

r_{max} – Maximum chest movement

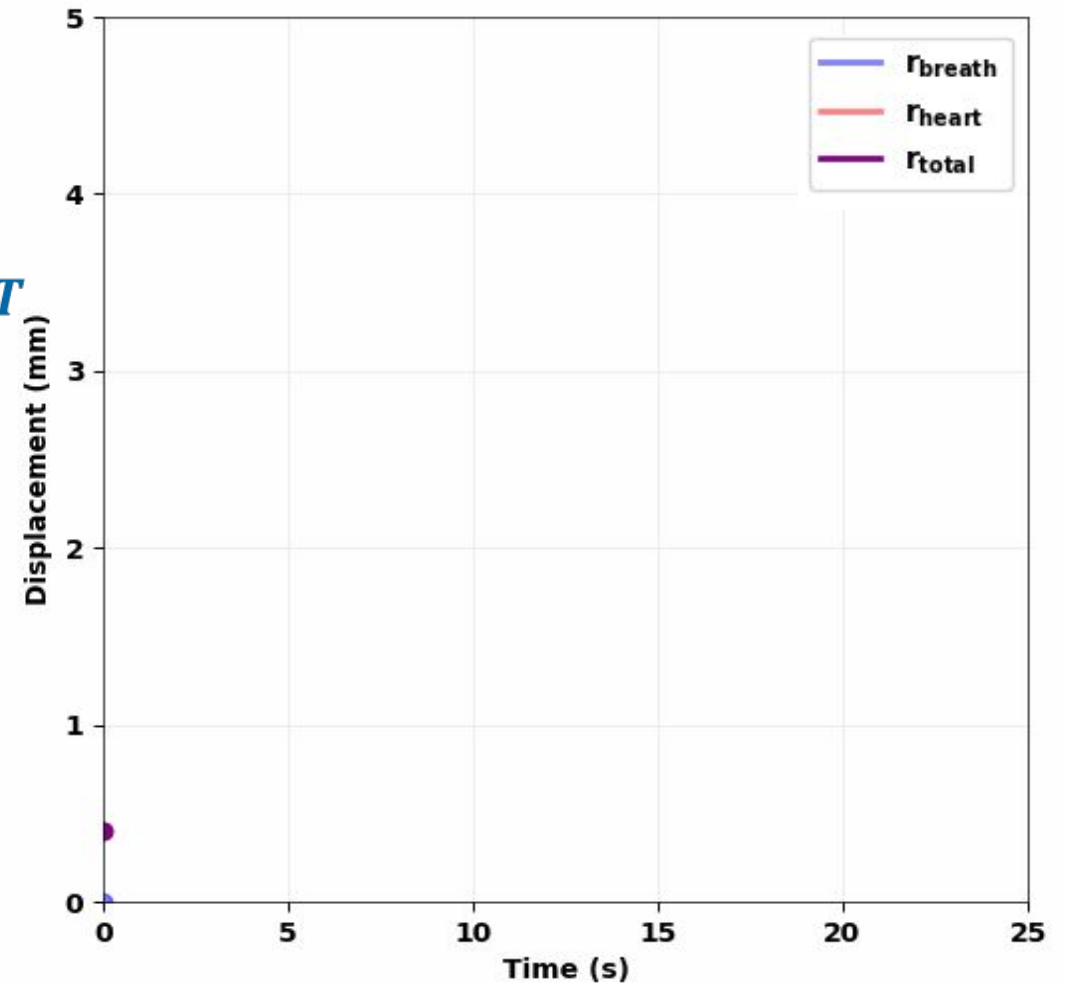
T_e – Expiration Time

τ – Decay Constant

$T = T_e + T_i$ – Time period of one breath

Heart rate estimated as sinusoid

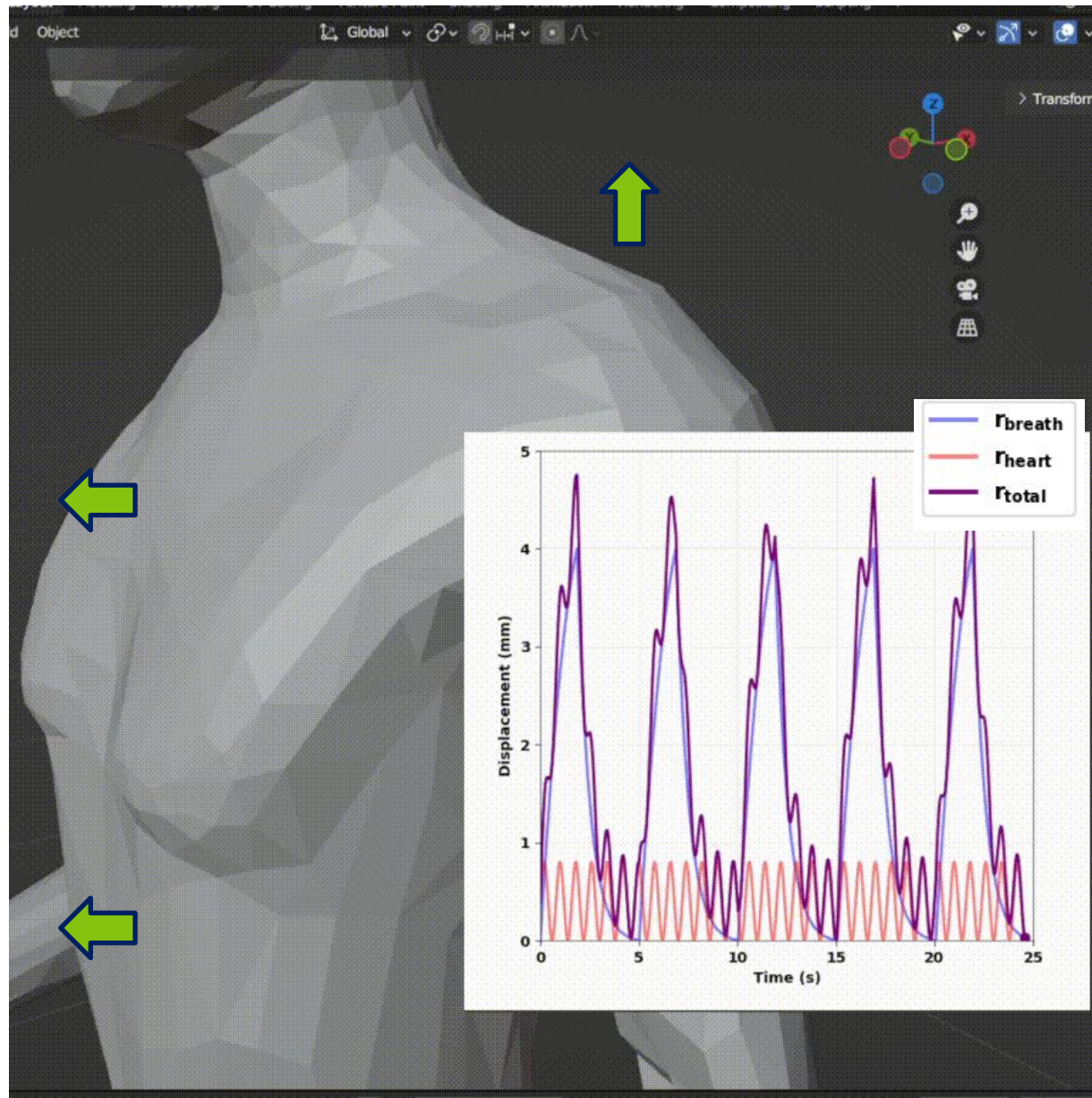
A. Albanese, L. Cheng, M. Ursino, and N. Chbat, "An integrated mathematical model of the human cardiopulmonary system: model development," Am. J Physiol Heart Circ Physiol, 2016.



Adding Motion With Blender

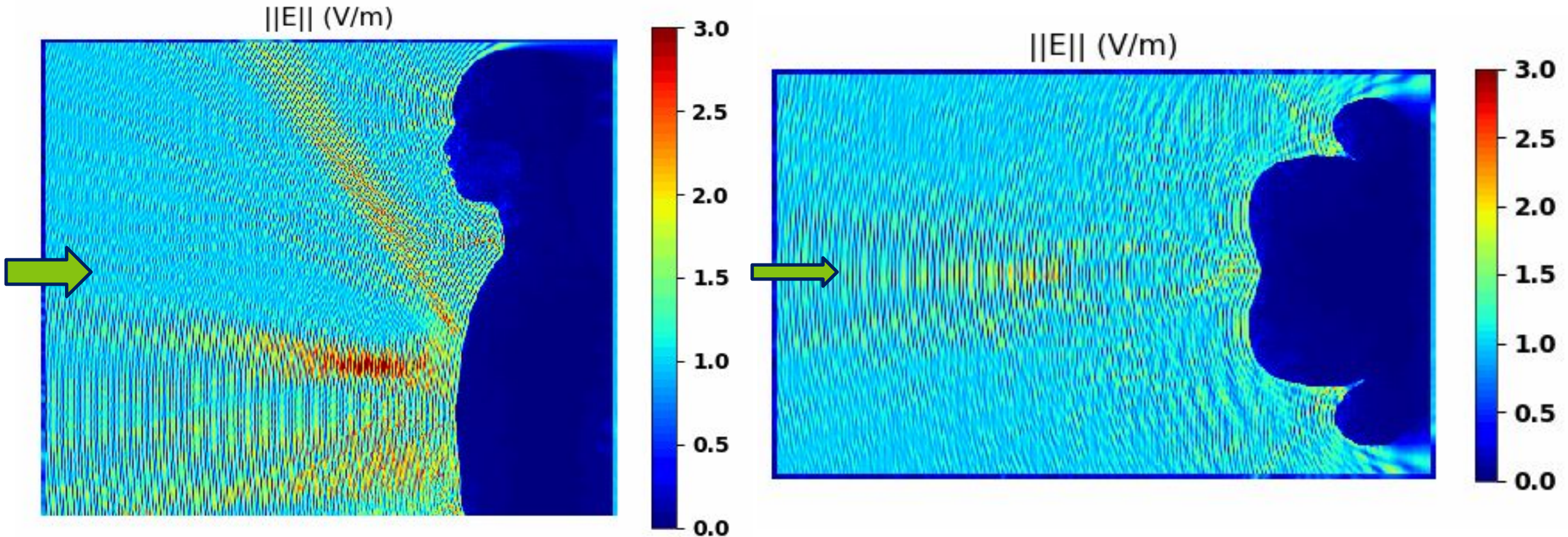
Explicit control of:
heart rate
breath rate
chest displacement

The entire breathing waveform



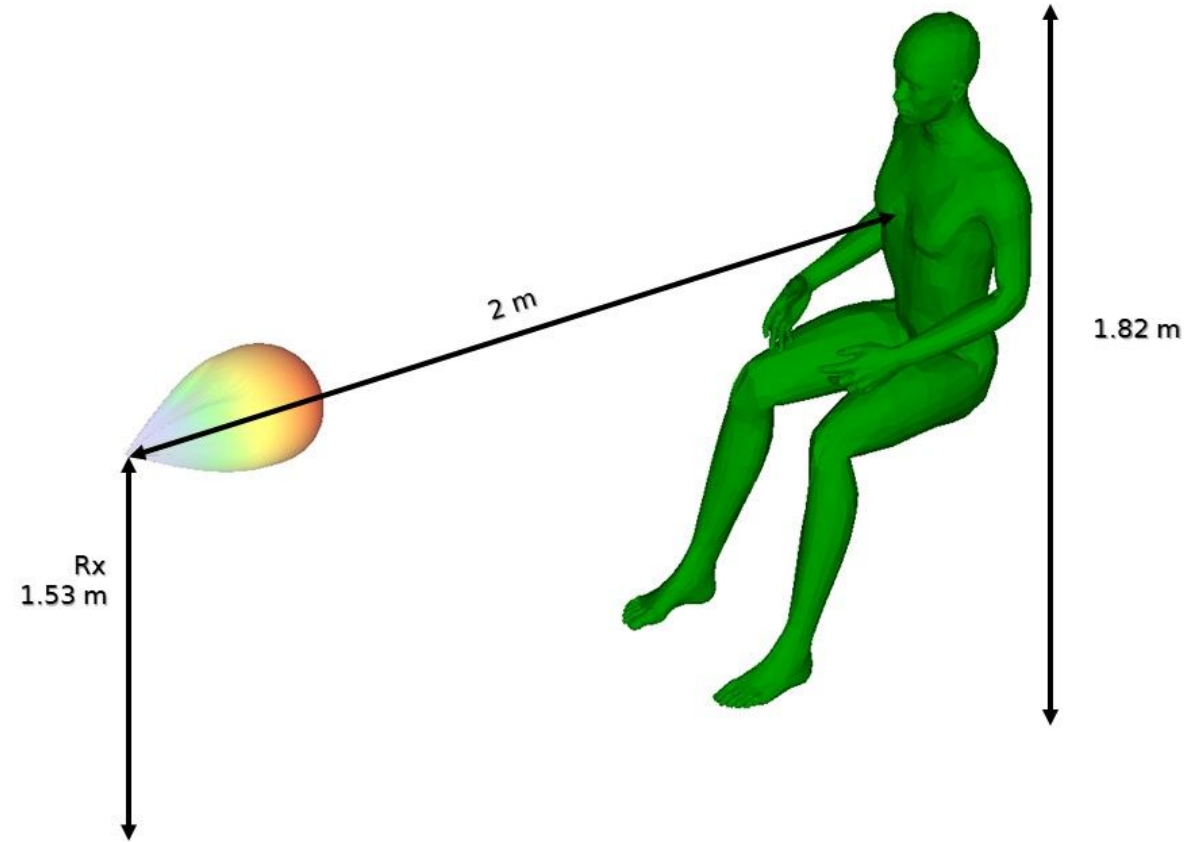
Full-Wave Simulation

28.5 GHz Plane Wave
2.4 s breath/ 1 frame every 50 ms = 48 sims
2 NVIDIA K80s, 30 min sim time * 48 sims = 24 hrs
2 mm x 2 mm x 2 mm resolution
30 GB GPU RAM



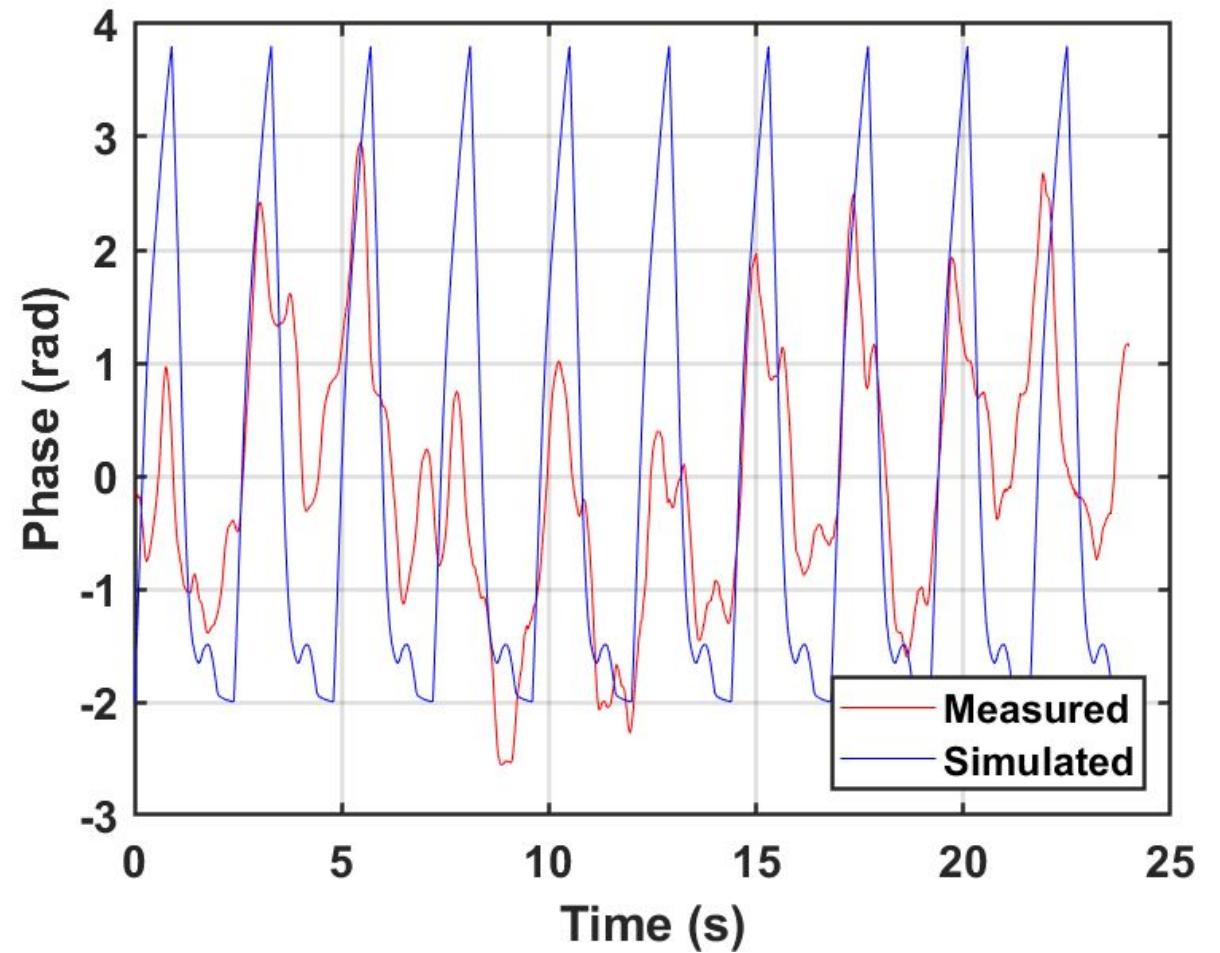
Ray Tracing Simulations

- 1 breath, sampled every 50 ms, 2.4 s breath
- $r_{\max} \approx 3.5 \text{ mm}$, $r_{h,\max} = 0.8 \text{ mm}$
- Human skin, $\epsilon_r = 15.5 - 14.2i$, for 28 GHz
- 22.5 deg HPBW in elevation, 30 deg HPBW in azimuth
- 2 min simulation time, 1.6 hours total.

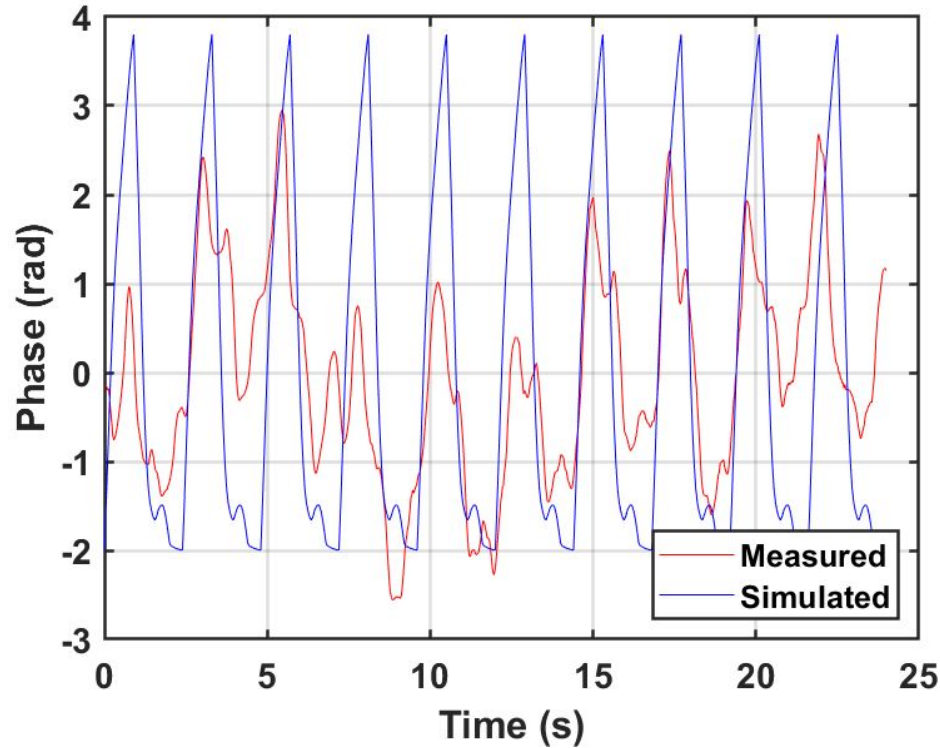


Results

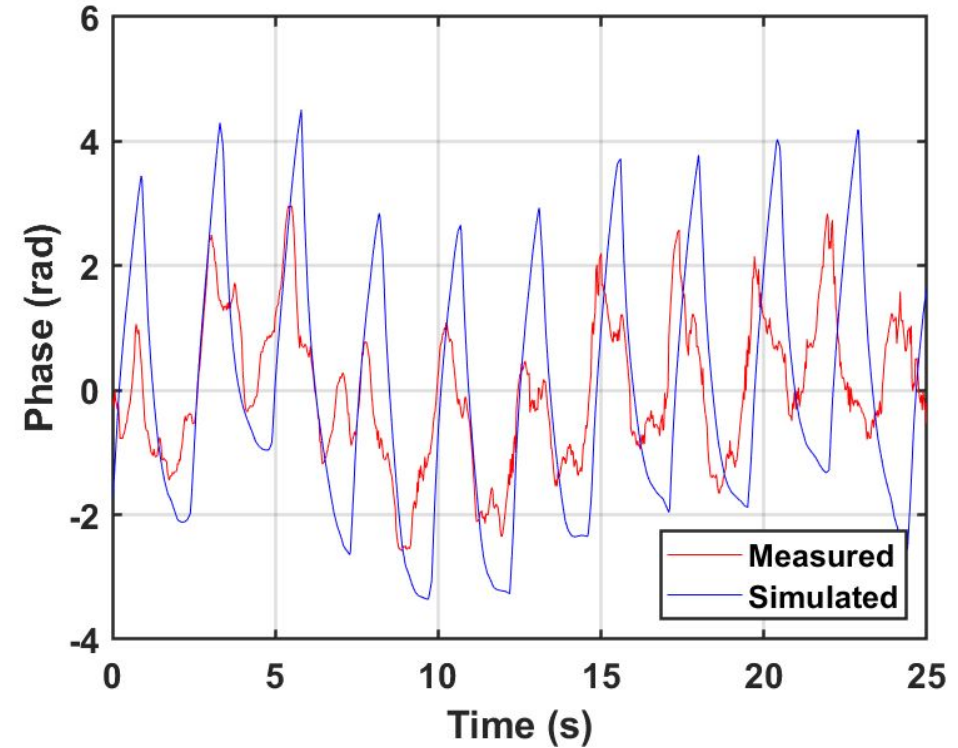
- 1 breath simulated 10x's
- No bias or movement correction
- Clearly motion and/or $r_{\max}$ varies with time



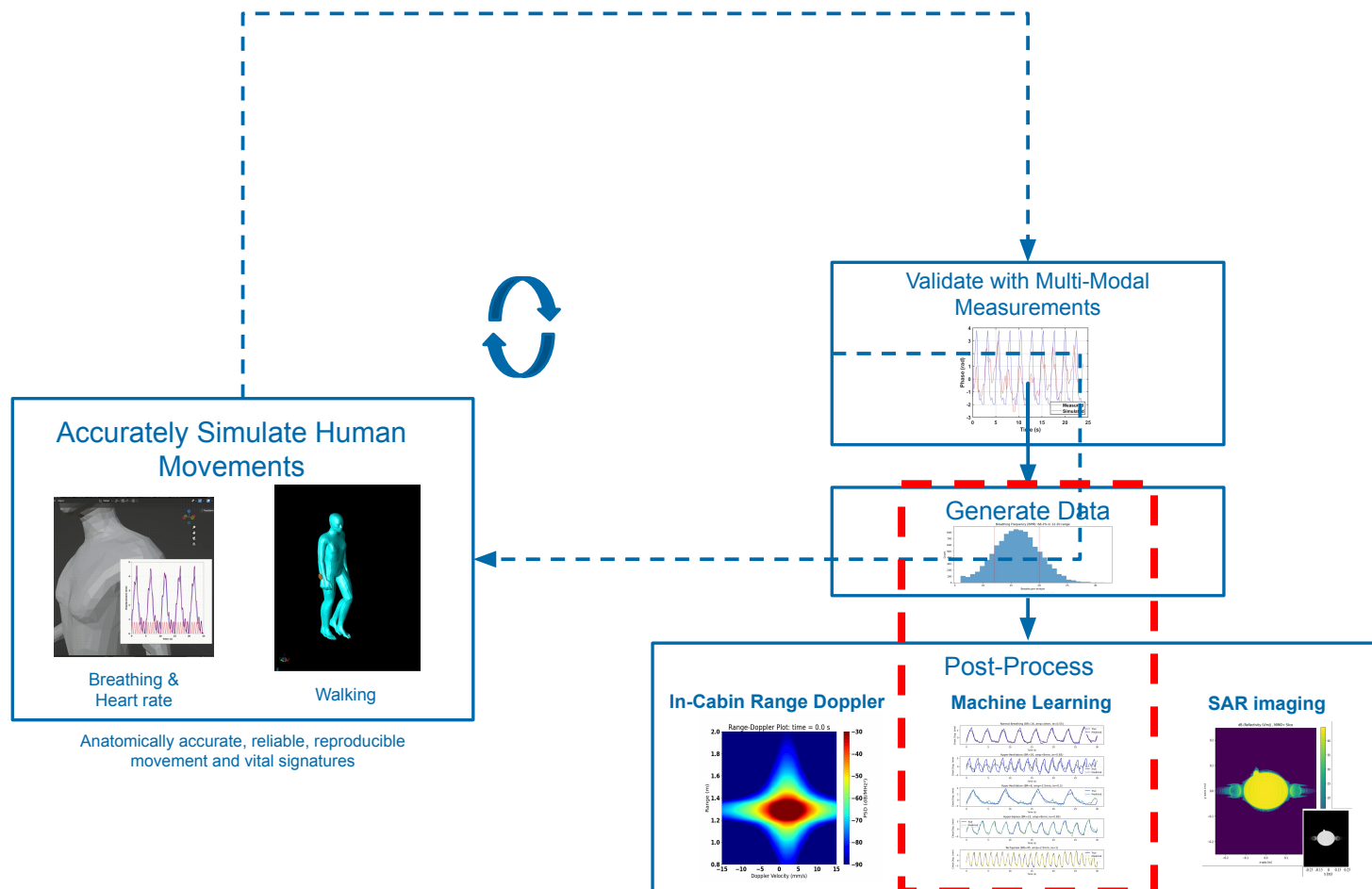
Phase Comparison with and without Bias



Without Bias

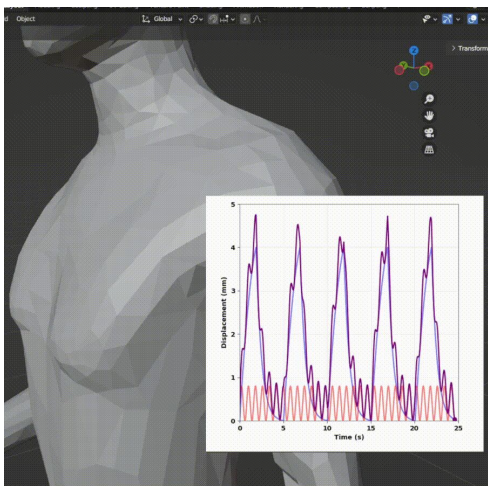


With Bias from
Lidar Data



With a validated breathing model we can begin generating mass amounts of data for a machine learning application.

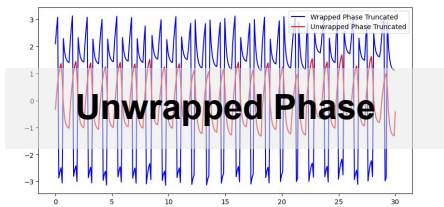
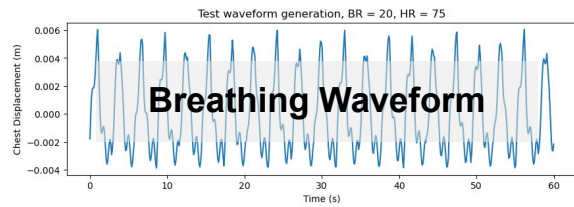
Predicting Breathing Patterns



- Automatically generate training, validation, testing data from breathing data
- Feed Convolutional Neural Network (CNN) the breathing waveform and the returned phase.
- Predict breathing waveform from the reflected phase

Parameter	Normal Breathing	Hyperventilation	Hypoventilation	Hyper-Apnea	Tachypnea
Frequency	12-20 BPM	20-40 BPM	6-10 BPM	15-30 BPM	30-60 BPM
Amplitude	~4 mm	~8 mm	~2-3 mm	~8 mm	~2-3 mm
Inhale-Exhale (IE) Ratio	0.5-0.6	0.7-1	0.25-0.33	0.7-1	~1

Neural Network Architecture



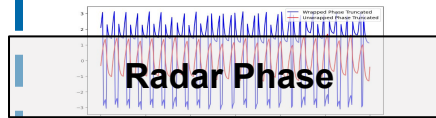
PO Based
Ray Tracing

x 200
waveforms



Ground Truth

70% Training
15% Testing
15% Validation

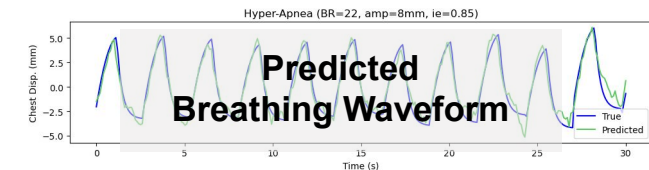


1D Convolution
filter = 32
kernel = 3

Max pooling
pool size = 2

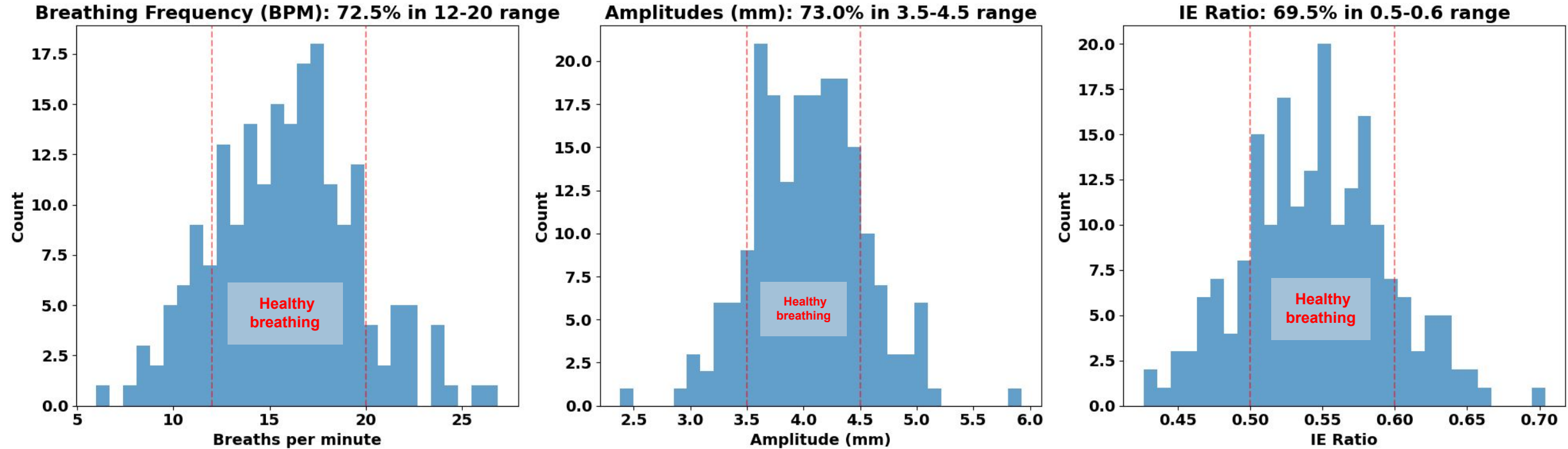
1D Convolution
filter = 64
kernel = 3

Linearization +
Dense Layer, 128 units



CNN Architecture

Input Synthetic Data - for training, testing, and validation

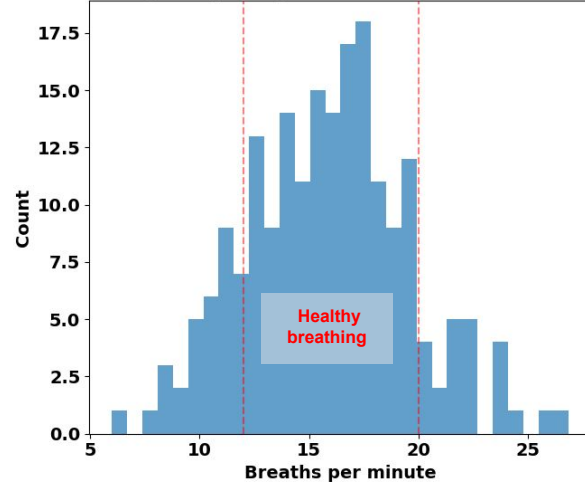


- Vary amplitude, IE ratio, and breathing frequency
- Add slight drift (mimics human swaying) and amplitude of breath variation
- Do we use a uniform or normal distribution?

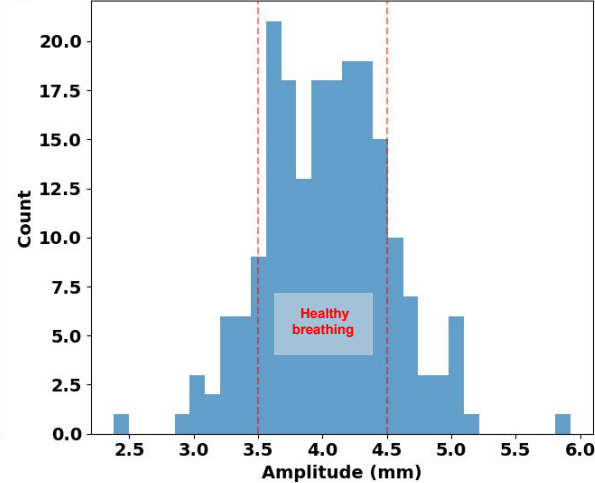
Uniform vs Normal Distribution

Normal
distribution
about healthy
parameters

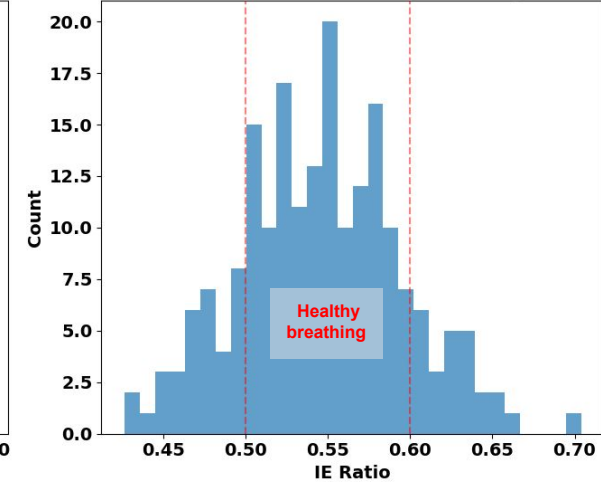
Breathing Frequency (BPM): 72.5% in 12-20 range



Amplitudes (mm): 73.0% in 3.5-4.5 range

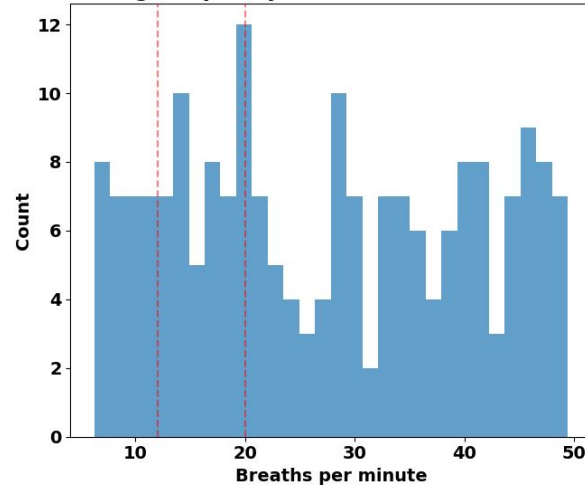


IE Ratio: 69.5% in 0.5-0.6 range

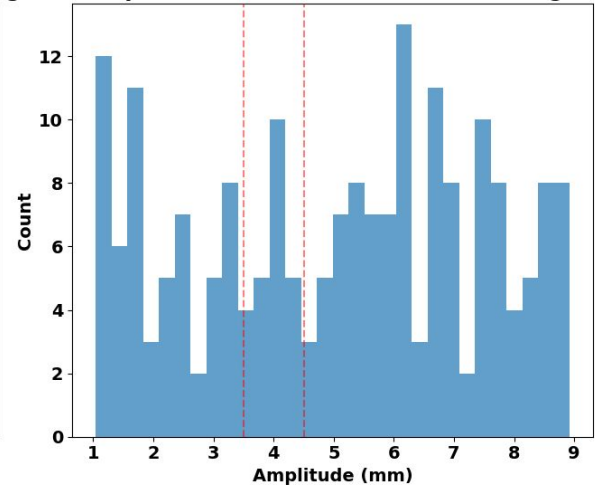


Uniform
distribution

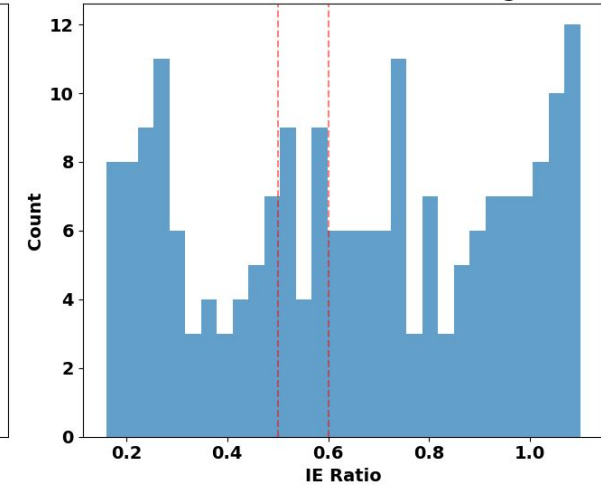
Breathing Frequency (BPM): 22.0% in 12-20 range



Amplitudes (mm): 12.5% in 3.5-4.5 range

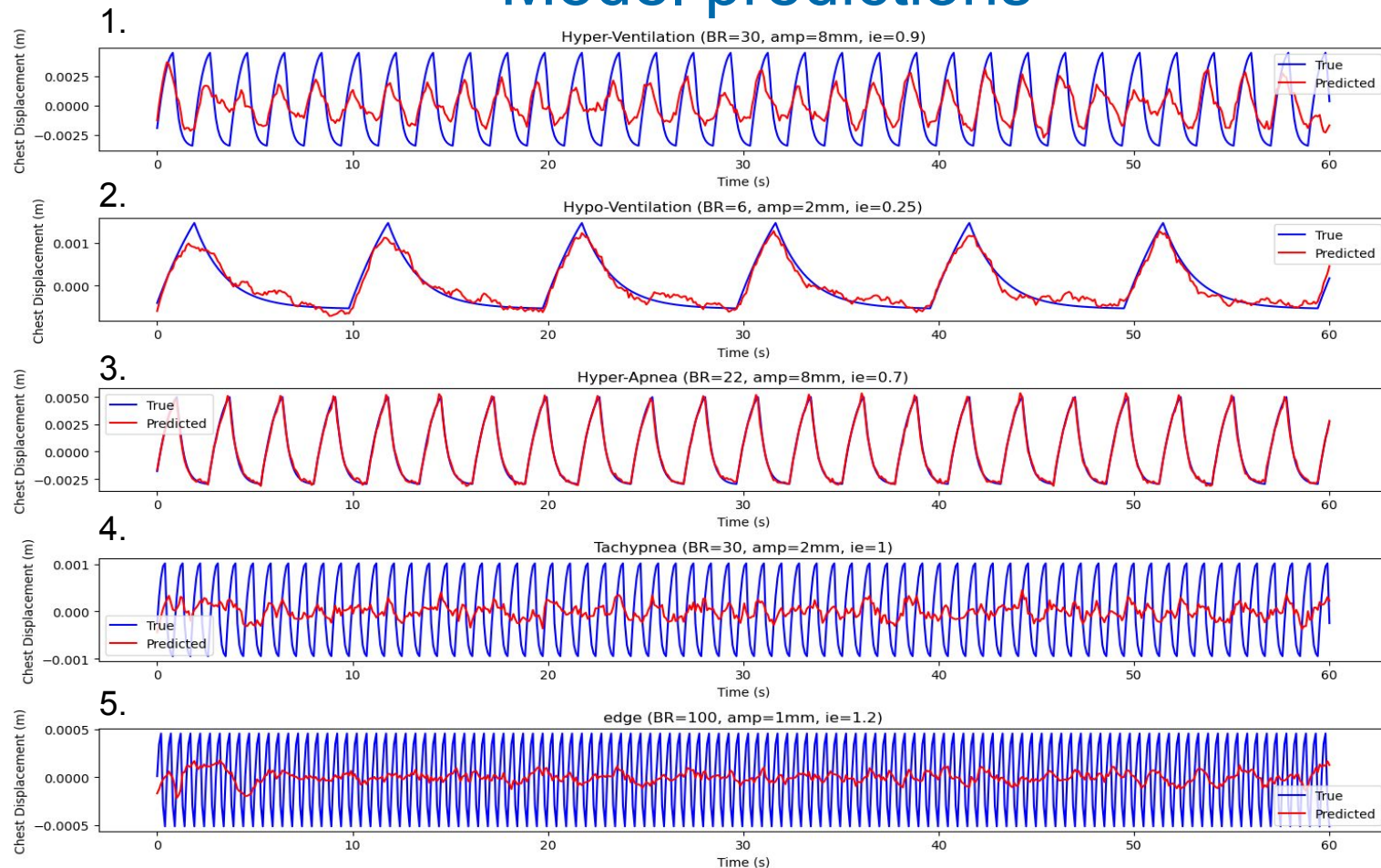


IE Ratio: 11.5% in 0.5-0.6 range

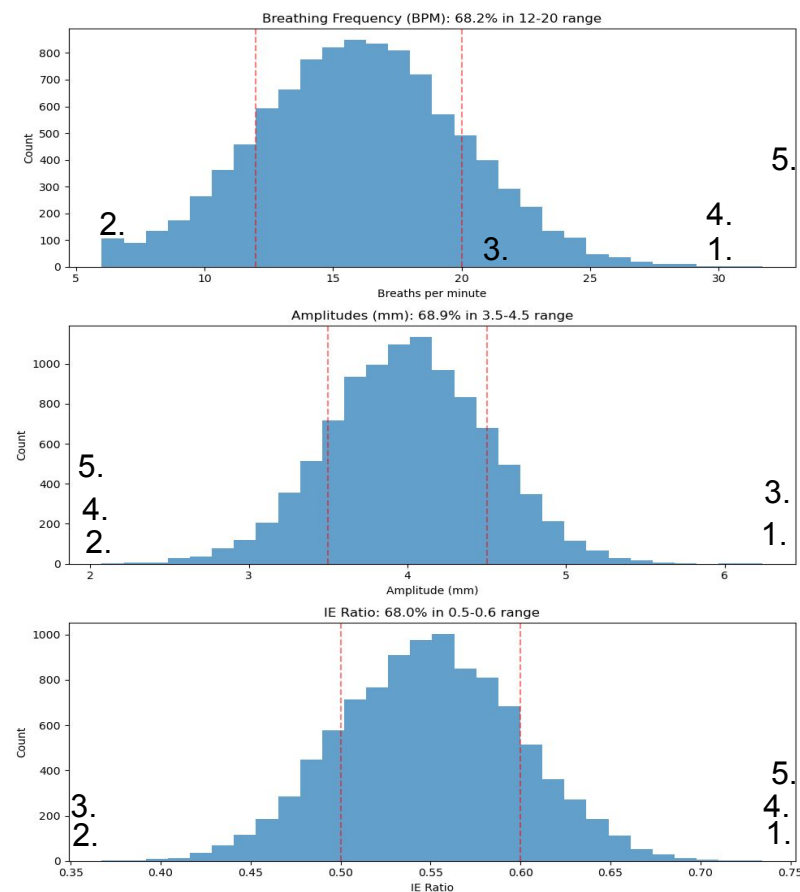


Normal Distribution - 10,000 waveforms

Model predictions



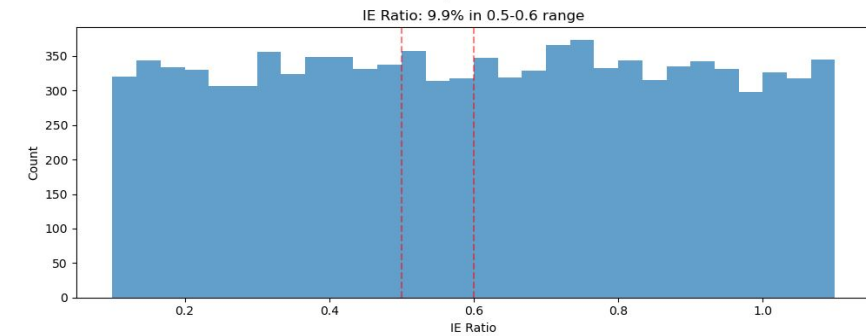
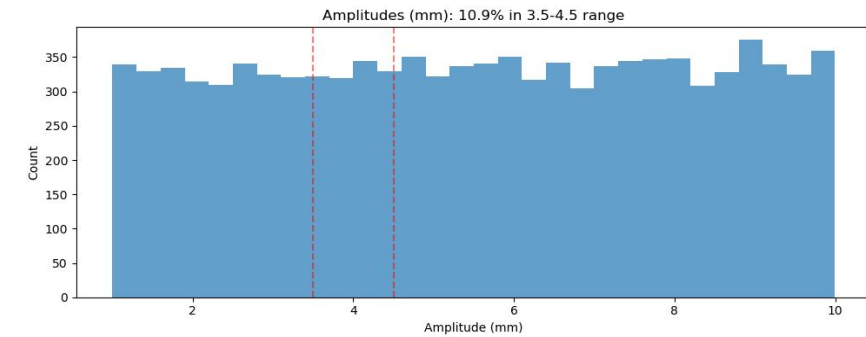
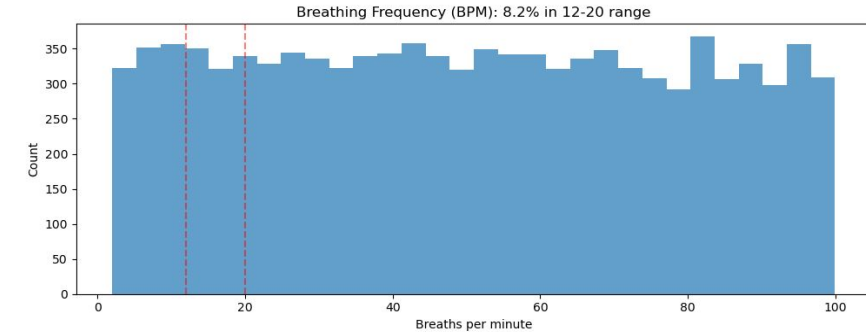
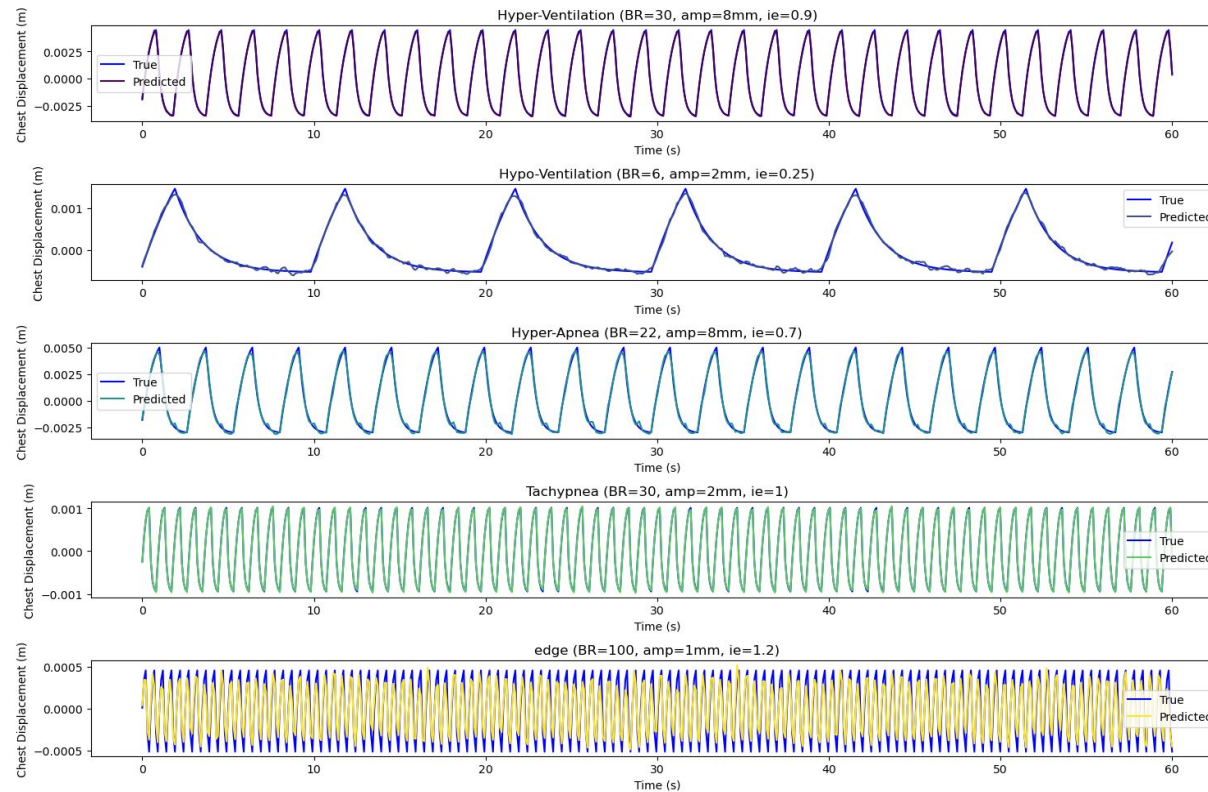
Synthetic Data



Uniform Distribution - 10,000 waveforms

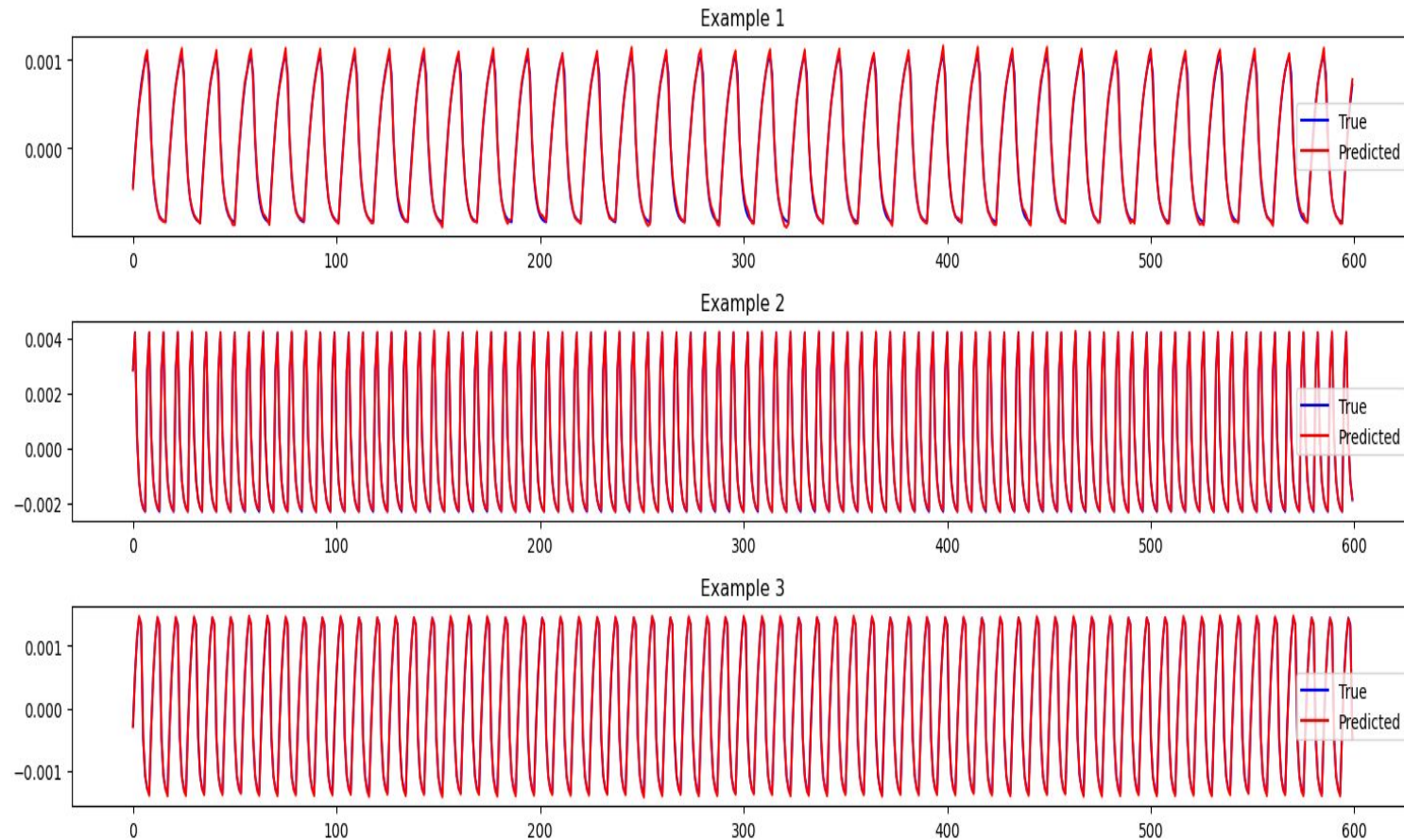
Synthetic Data

Model predictions



Uniform vs Normal Distribution

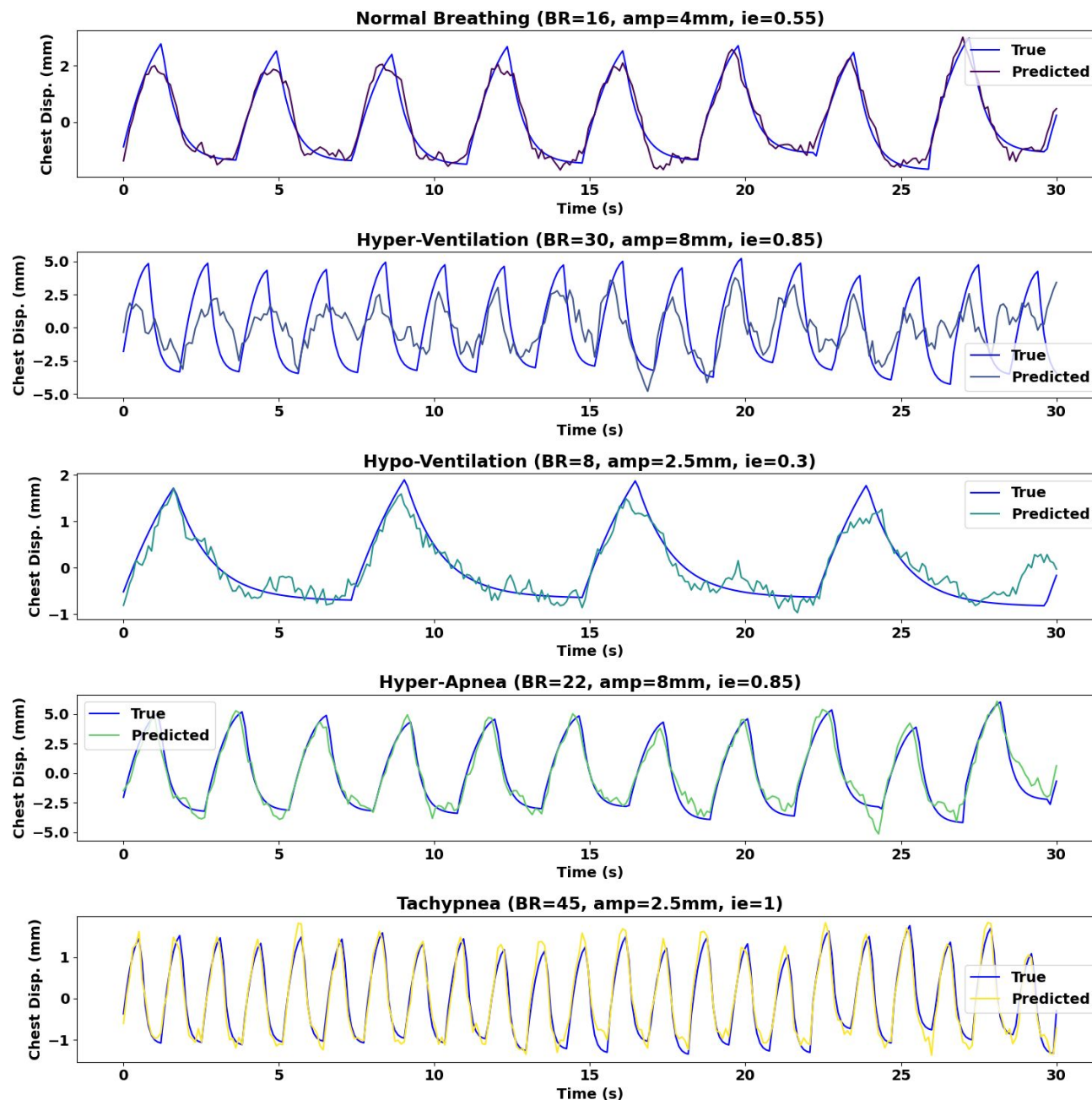
- With uniform distributed dataset model performs much better.
- Simulation offers easy generation of edge cases (e.g 50 breaths/min) and thus the model performs much better predicting unhealthy breathing!



*200 waveforms used to avoid overfitting

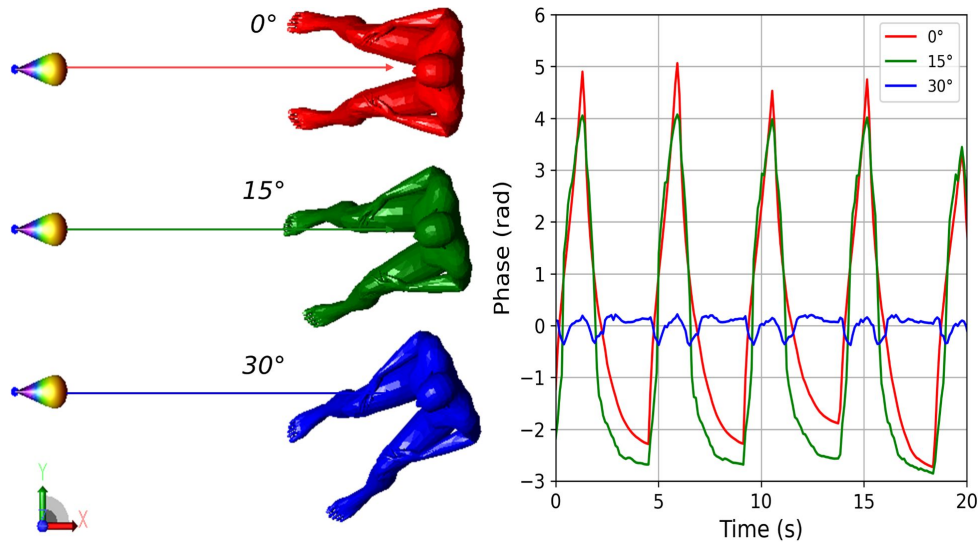
Results

- Training MSE = 2.6%
- Validation MSE = 9.4%
- Root mean square error (RMSE) –
Normal = 0.3 mm
hyper-vent. = 2.2 mm
hypo-vent. = 0.3 mm
hyper-apnea = 0.7 mm
tachypnea = 0.2 mm
- Hyper-Ventilation suffered most since variables were towards the edge of training data set

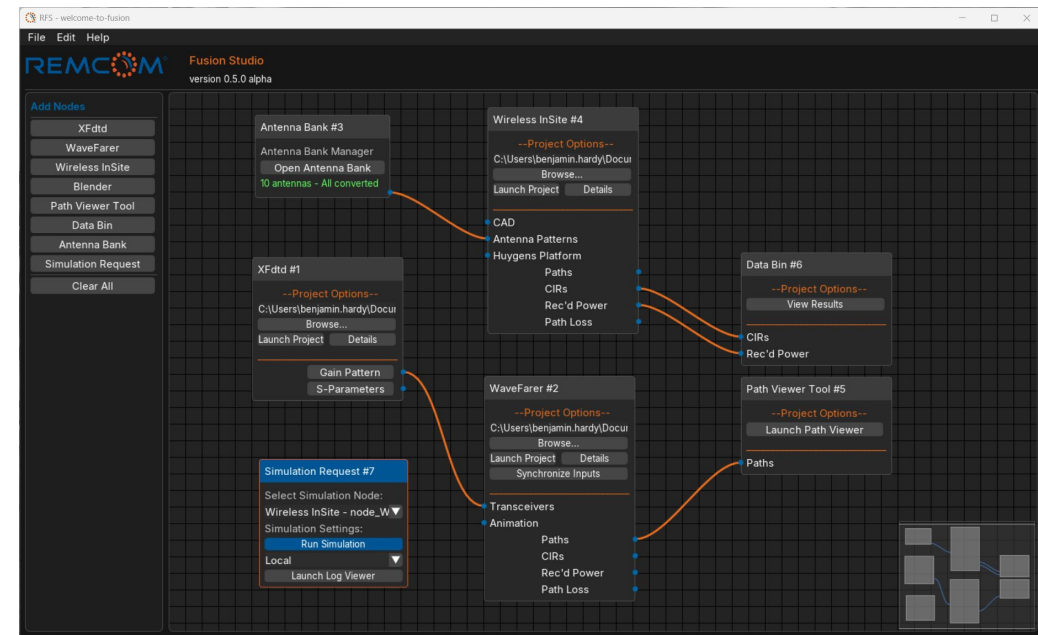


Future Direction

Further human and scene modification adding rotation, furniture, clothing, etc.

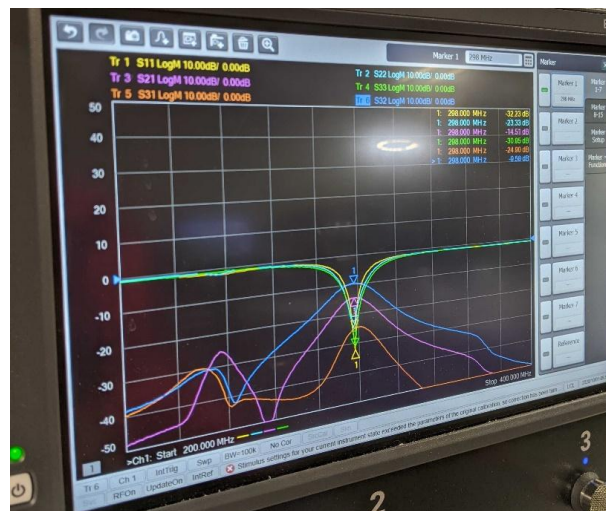
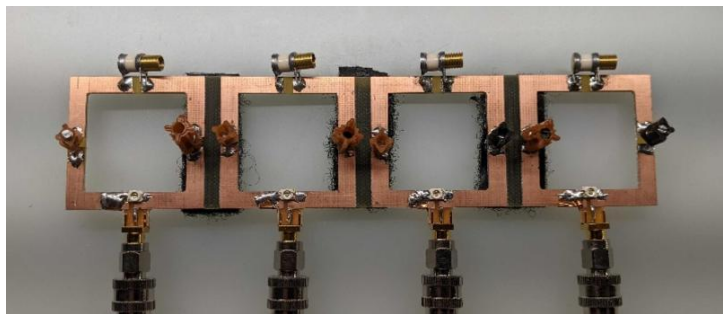


Automation built into a new tool called
Remcom Fusion Studio (RFS)

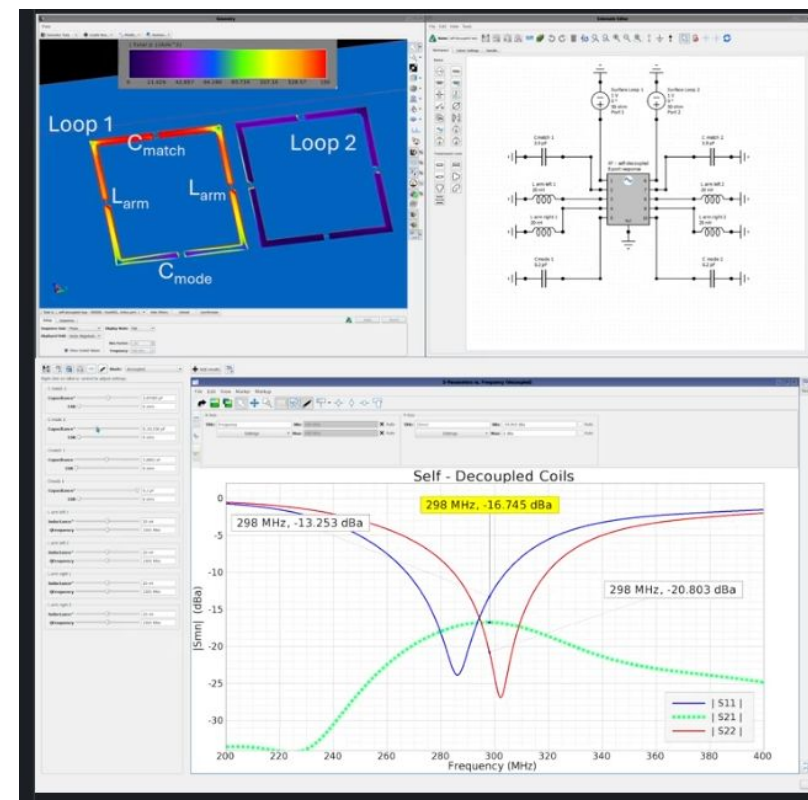
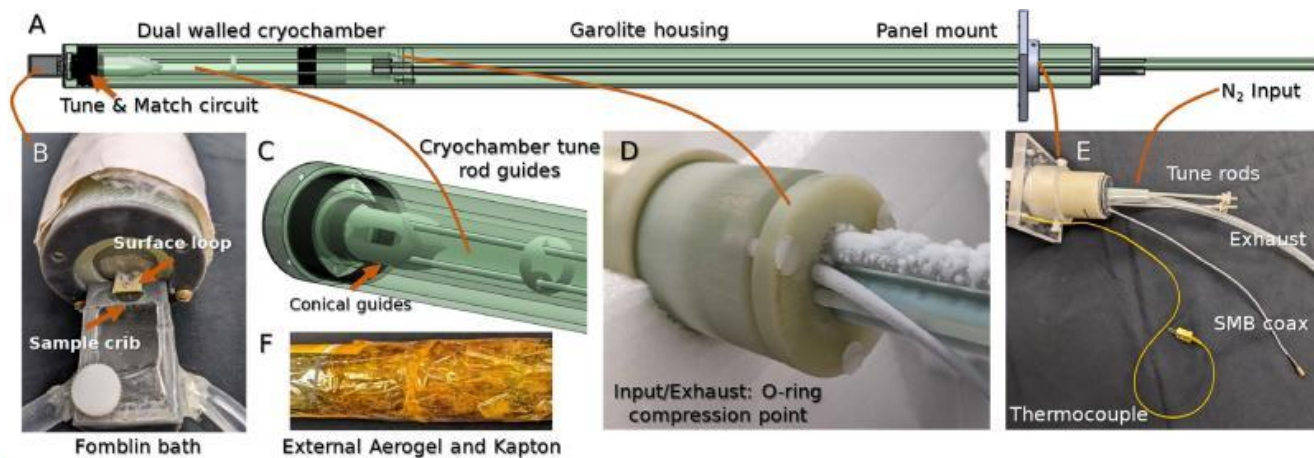




My background



7T MRI (298 MHz) Array hardware and simulation



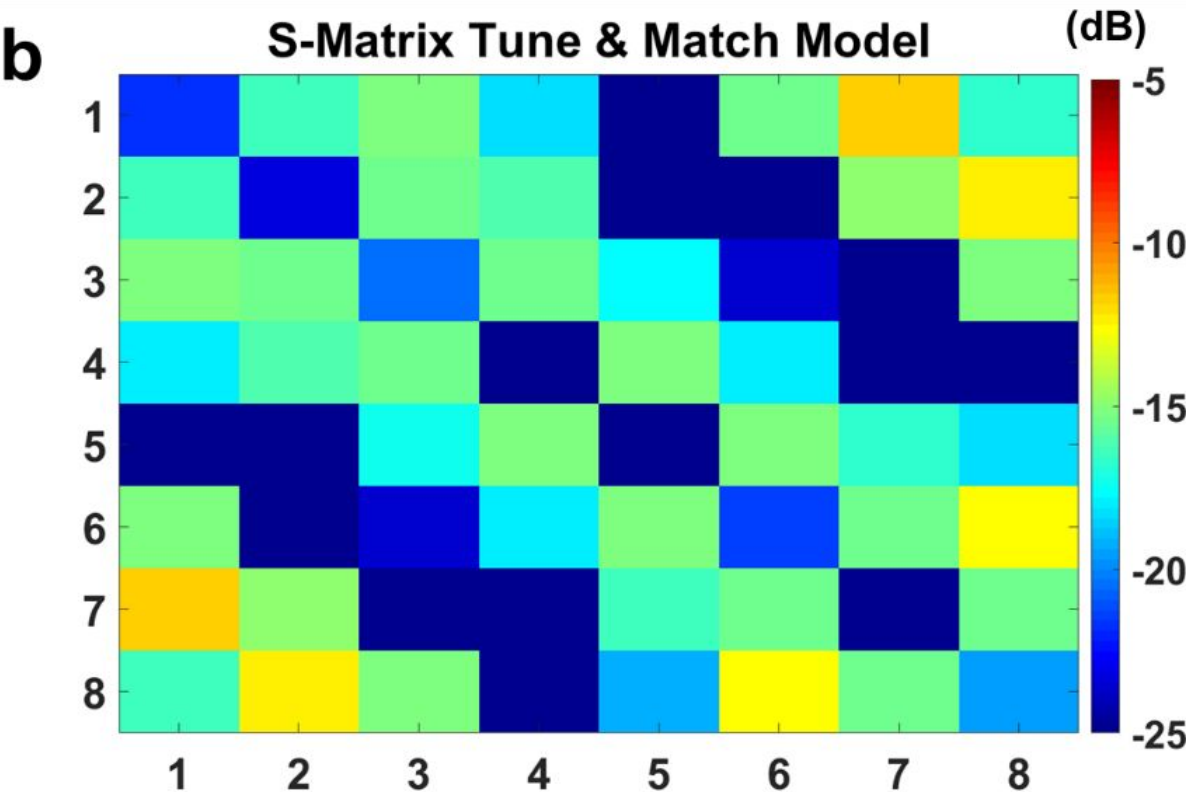
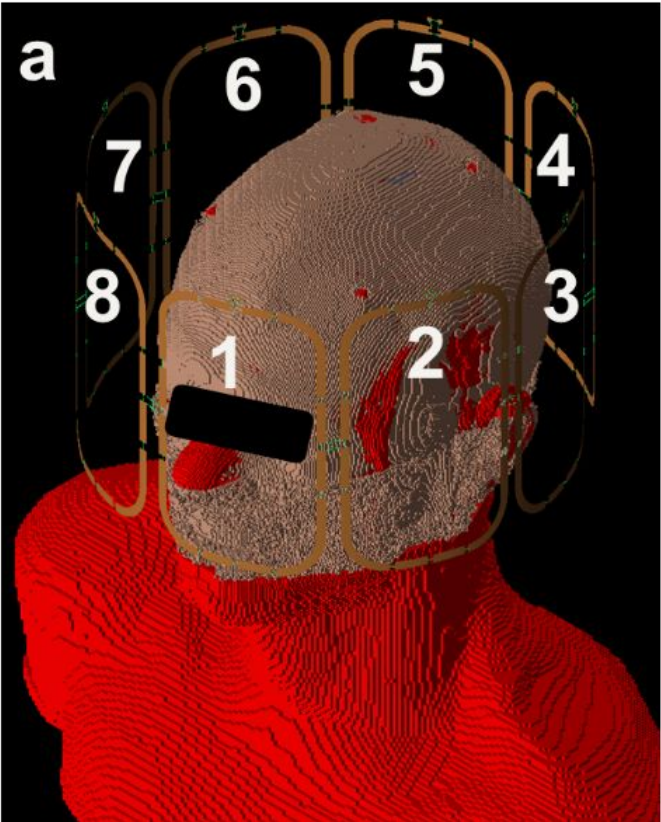
15.2T (650 MHz) Cryogenic RF probe

7T MRI 8 channel Head Coil

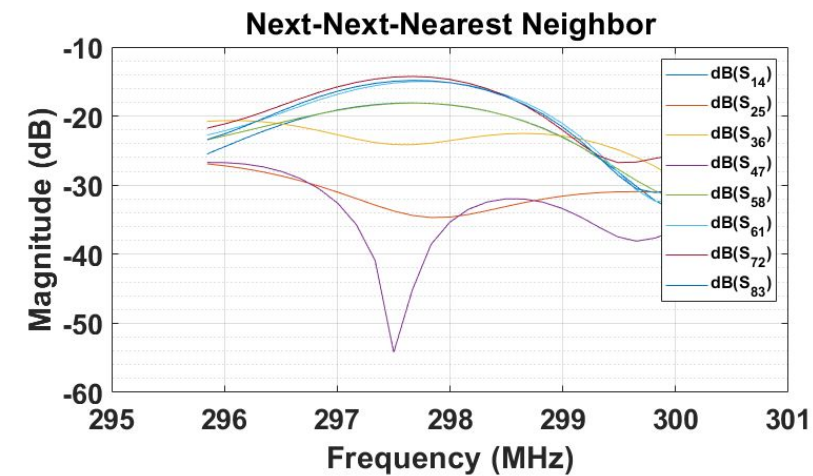
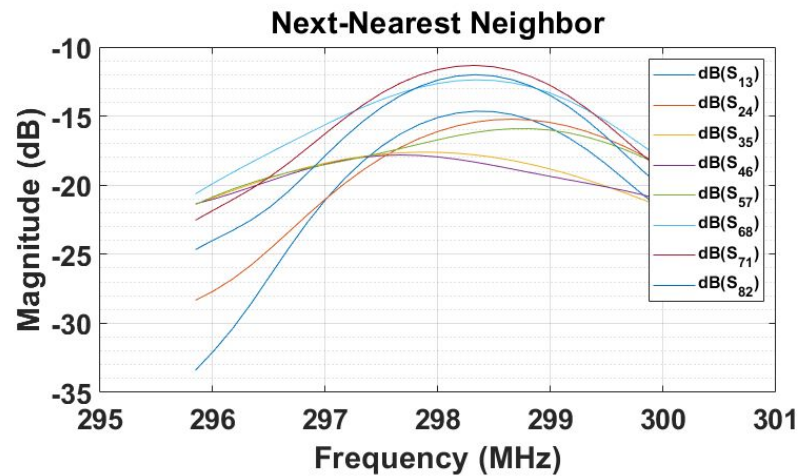
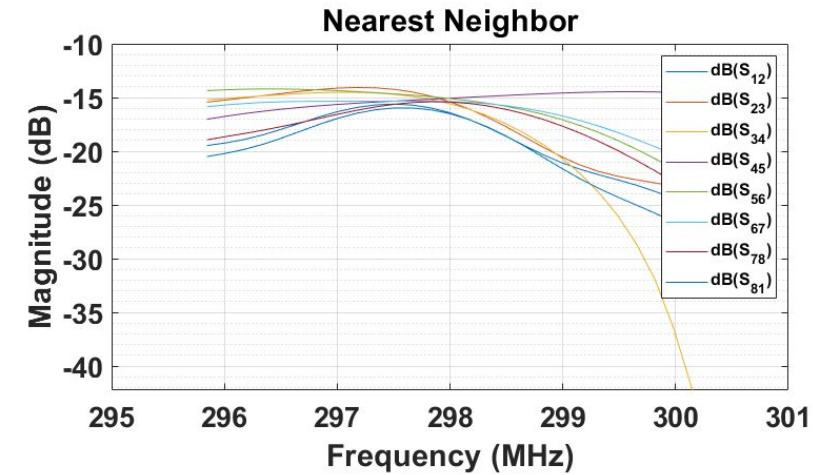
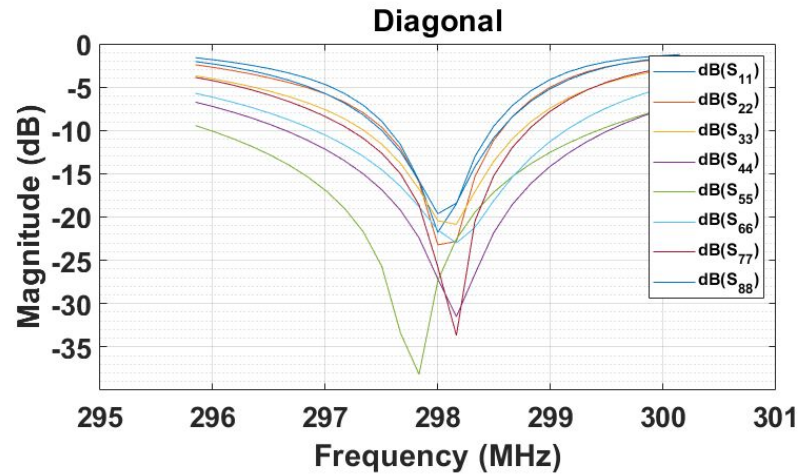
8 channel MRI head coil

Simulated 72 heads for Specific Absorption Rate variability

Used ADS for 96 port circuit optimization

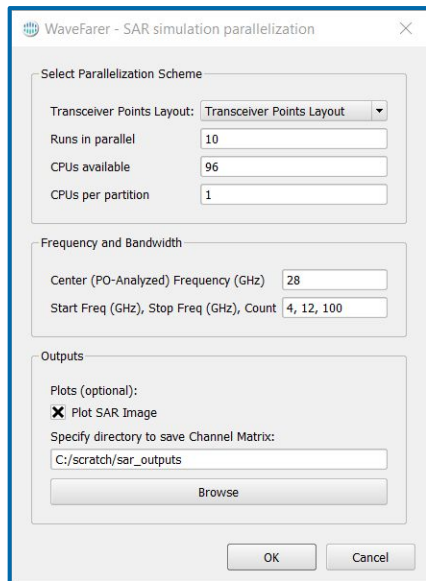


Decoupled using ADS



SAR Imaging Application

User Inputs and Project Setup



WaveFarer - SAR simulation parallelization

Select Parallelization Scheme

Transceiver Points Layout:

Runs in parallel:

CPUs available:

CPUs per partition:

Frequency and Bandwidth

Center (PO-Analyzed) Frequency (GHz):

Start Freq (GHz), Stop Freq (GHz), Count:

Outputs

Plots (optional):

☒ Plot SAR Image

Specify directory to save Channel Matrix:

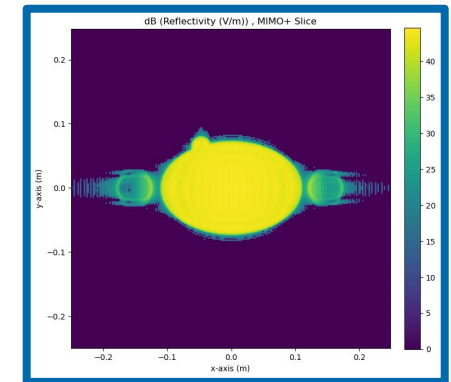
PO based Ray Tracing
Center Frequency
Parallelizable down to each Tx

**Complex Impulse Response
Generator**

Channel Matrix
E (frequency, Rx, Tx)

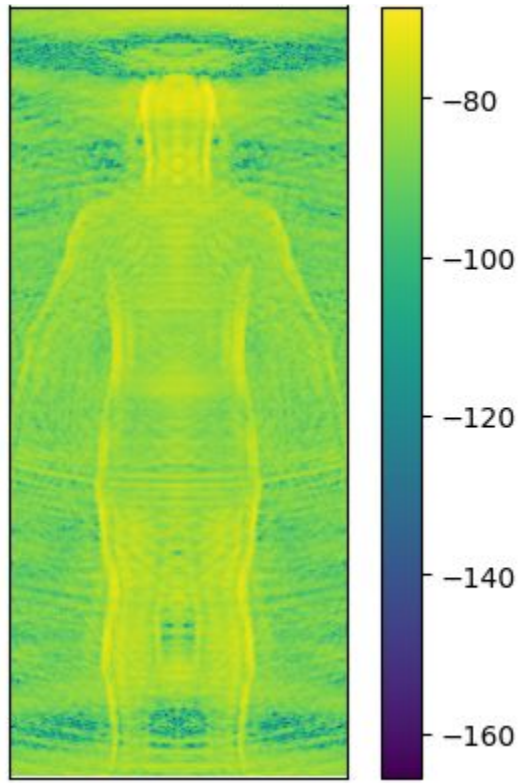
Broadband Analyzer

Reconstruction



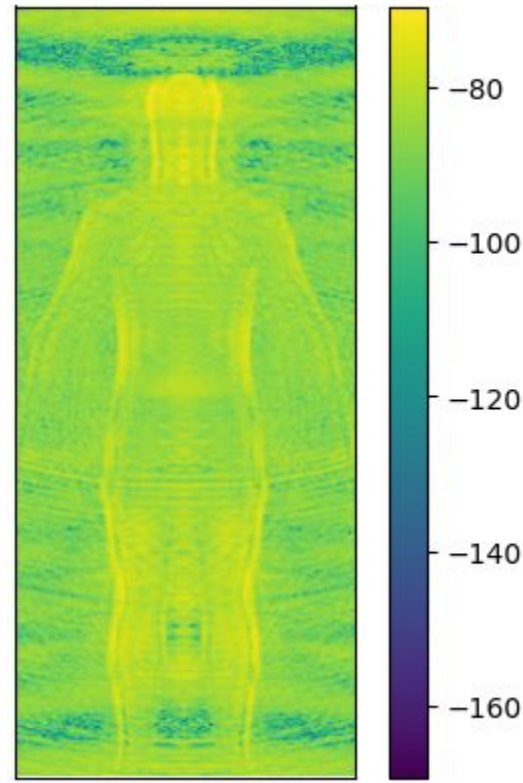
Human Imaging – 100s of Tx/Rx combos

Ray tracing for
30 frequencies

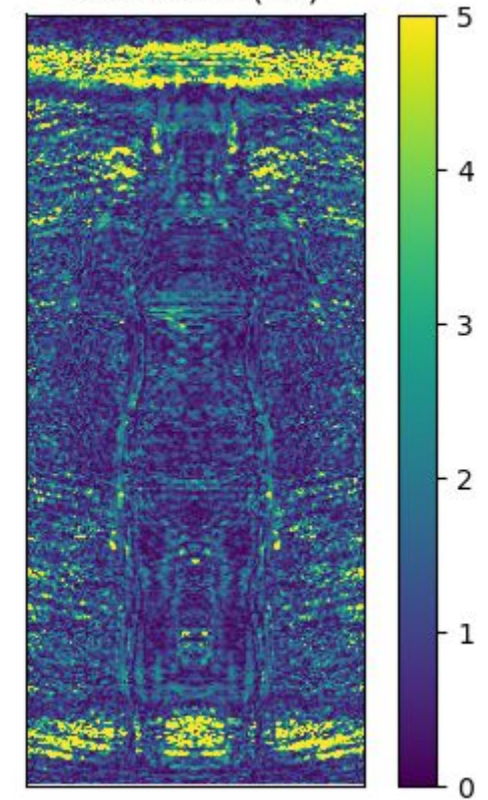


Broadband
Workflow

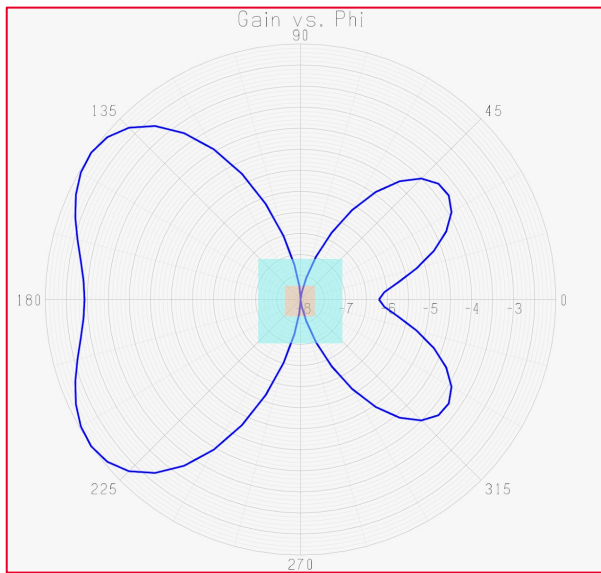
3x speedup



Difference (dB)



Other Applications



Geometry Overlay Tool on
2D gain plots

XFdd - Antenna Array Analyzer [pid:24836]

Antenna Array Analyzer

Array Configuration

Array Type: Waveguide Linear Array

Array Dimensions

Elements X: 4
Elements Y: 4
Elements Z: 1
Spacing X: 6.9 mm
Spacing Y: 8.2 mm
Spacing Z: 0 mm

Geometry and Component Duplication

Number of Components/Waveguides: 2

☒ Duplicate Selected Geometry
☒ Duplicate Circuit Components/Waveguides

Duplicate Selected Items

Note: Select geometry assemblies and/or circuit components/waveguides first

Simulation Control

☒ Run Simulation After Creation
☒ Extract S-Parameters and Gain Patterns

Create and Run Simulation

Python Post-Processing

Python Executable: python

☒ Launch Python Analysis UI

Analysis Options:

☒ Calculate Array Factor
☒ Mutual Coupling Analysis
☒ Beamforming Analysis
☒ Scan Blindness Prediction
☒ Grating Lobe Suppression
☒ Polarization Analysis

Output Settings:

Output Directory: gents/single-element.xf

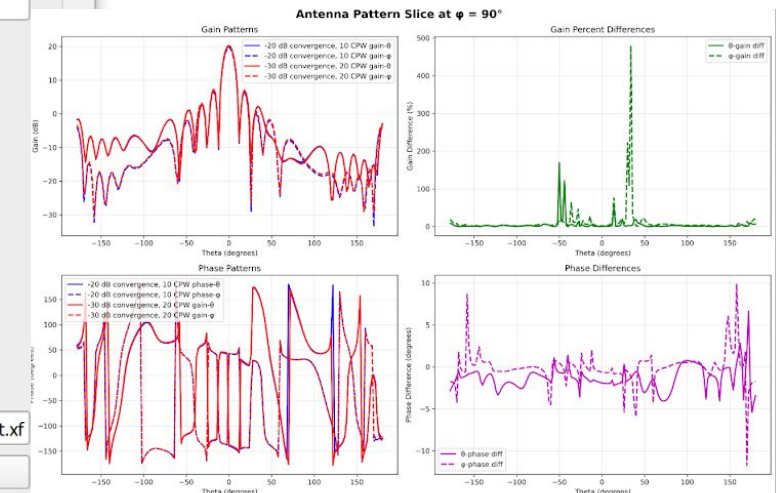
Browse...

Launch Python Analysis

OK Cancel



Phased Array Builder and Post-processing Toolbox



Accelerating Solutions with Agents

Using Cursor agents I have accelerated (days → hours) several automation routines and UI's for customers including

- Local SAR calculation/parallelization routine (10x speed up)
- Python based Simulation Node Editor that connects all of Remcom's tools
- Customized data plotting inside XFDTD
- UI for bulk data export





Thank you!
Questions